

“Workhorses of Opportunity”: Regional Universities Increase Local Social Mobility

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 - ▶ Disproportionately from lower-income families and racial minorities
- ▶ Serving nearby students was important justification for these univ.
- ▶ This paper: new strategy to identify effects of these universities on economic mobility of nearby children
 - ▶ Placement of normal (teacher's schools) vs. insane asylums in late 19th/early 20th century
 - ▶ Normal schools converted to regional universities
 - ▶ Asylums are now mostly psychiatric facilities, and stayed small in size

Preview of Findings

- ▶ Regional universities increase education, labor market, and social outcomes of nearby children
 - ▶ In early 1900s and today
 - ▶ Especially children in low-income families
 - ▶ Causal effect on children, not just selection
- ▶ Examine mechanisms using “The Freshman Survey”
 - ▶ Proximity ↓ financial cost of college and ↑ information
 - ▶ Results not driven by universities’ effects on the local economy

Today: New results on HBCUs (joint with Namgyoon Oh)

- ▶ Especially interesting is the effect of HBCUs, some of which also began as normal schools
 - ▶ Focus on access for particular demographic group
- ▶ New identification assumption: same-state counties with an exclusively Black asylum are a good control group for counties with a Black normal school
- ▶ Look separately at the effects on Black and white children

Normal School History

- ▶ Mid-19th century begins an era of government-chartered normal schools to train teachers
 - ▶ Part of a broader social movement to increase public institutions for general well-being
- ▶ 209 normal schools opened from 1839-1930 (Ogren 2005)
 - ▶ Often multiple established within a state

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- ▶ 209 normal schools opened from 1839-1930 (Ogren 2005)
 - ▶ Often multiple established within a state
- ▶ Established through legislation; location chosen by legislature or commissions established by the legislature
- ▶ Factors determining locations of normal schools (Humphreys 1923):
 - ▶ Political influence
 - ▶ Demand for instruction
 - ▶ Accessibility
 - ▶ Financial and Land Donations
 - ▶ Locations of Existing Schools
 - ▶ Natural Beauty

Normal School History

DeKalb chosen as location of new normal school in northern Illinois (1895)



A surviving piece of ephemera celebrating the choice of DeKalb for the new state normal school, this embroidered and painted cloth still resides in NIU's Regional History Center / Archives.

Source: Northern Illinois University website

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 - ▶ They already existed
 - ▶ They were geographically distributed across the state
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- ▶ 1950s: Many become state colleges
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 - ▶ 50 percent of public comprehensive univ. started as normal schools

Asylums: Similar time and criteria as Normal Schools



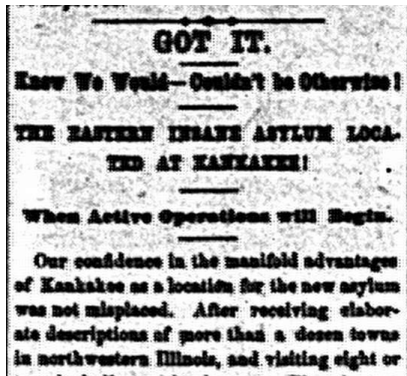
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 - ▶ Goal of restoring patients
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 - ▶ Goal of restoring patients
- ▶ Typically locations from legislatures, sometimes same day as normal schools
- ▶ Factors other than political influence (Kirkbride 1854):
 - ▶ Close to roads, railroads, population centers
 - ▶ Picturesque locations

Asylum desired by local communities



Kankakee Gazette, August 9, 1877

"Our citizens received the news in a spirit of jubilee, and on Friday evening there was a bonfire, band music... and speeches.."

- ▶ Deinstitutionalization movement: 1950-1980
- ▶ Today, many are psychiatric facilities, a few are prisons, some acquired by universities

Non-econometric evidence of balance

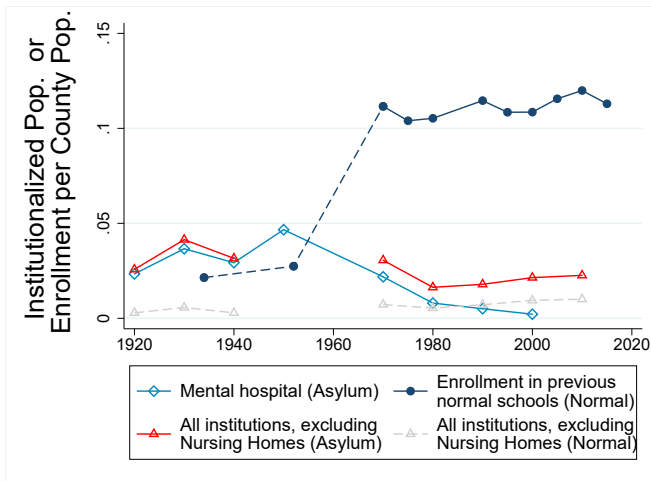


Non-econometric evidence of balance



Black-and-white are normal schools, color photos are asylums

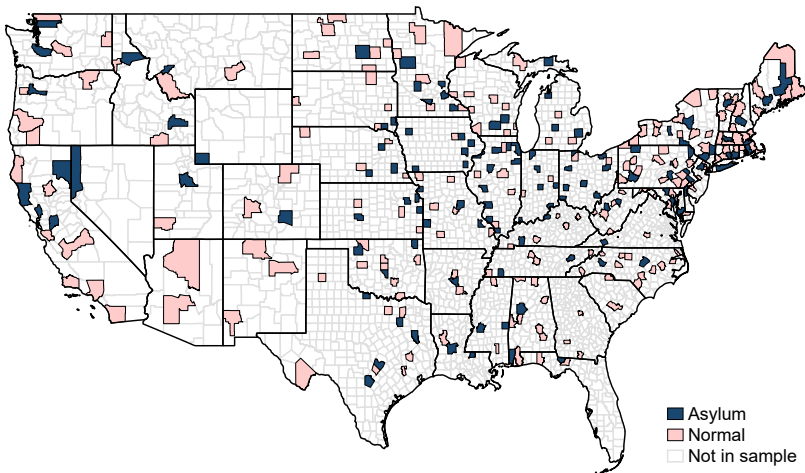
Timeline



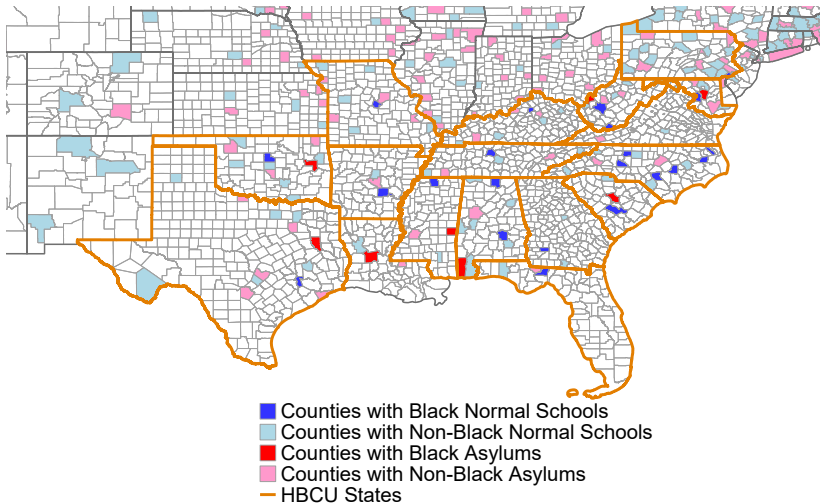
Data Collection: Normal Schools and Asylums

- ▶ Historical data on normal school location comes from Ogren (2005)
- ▶ We digitize data on asylums from the 1923 special census of “institutions of mental disease”
- ▶ 204 normal counties, 125 asylum counties

Map of Normal Schools and Insane Asylums



Map of Black Normal Schools and Black Asylums



Empirical Strategy

- ▶ Main specification:

$$y_i = \beta \text{Normal}_i + \alpha_s + \epsilon_i \quad (1)$$

- ▶ Sample: Normal school and Asylum counties
 - ▶ For HBCUs: Black normal schools and Black asylums
- ▶ Include state fixed effects
- ▶ Identification assumption: asylum counties in the same state are a good counterfactual for the social mobility of normal school counties
 - ▶ had the normal school county received a different state institution
- ▶ Standard errors clustered at the state level

Timeline for empirical analysis

1. Pre-period balance in mid 1800s

- ▶ **County characteristics:** full-count 1840 and 1850 census
- ▶ **Social mobility:** matched census records of 1850 children to 1860 adults (Census Tree)

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 - ▶ **Social mobility:** matched census records of 1920 children to 1940 adults
3. Effects of normal school assignment today
 - ▶ **County characteristics:** Census, BEA, Opportunity Insights
 - ▶ **Social Mobility:** 1978-1983 birth cohort outcomes as adults
 - ▶ **Mechanisms:** The Freshman Survey

Pre-period Balance in 1840

	(1)	(2)	(3)	(4)	(5)
	Variable Means			Difference in Means With State FE	
	Normal	Asylum	All others	(1) - (2)	(1) - (3)
Panel A: Geographic Characteristics					
Within 150 Miles of State Capital	0.48 (0.5)	0.52 (0.5)	0.42 (0.49)	-0.11* (0.06)	-0.02 (0.03)
Water Coverage	6.23 (12.96)	6.35 (12.44)	4.39 (11.06)	-0.04 (1.34)	-0.83 (0.92)
Panel B: Characteristics in 1840					
Log Population	9.71 (1.38)	9.51 (1.59)	8.99 (1.22)	-0.02 (0.13)	0.27*** (0.08)
Log Nearby Population	11.00 (0.36)	10.94 (0.37)	10.92 (0.33)	-0.02 (0.02)	0.001 (0.01)
Log Manufacturing Capital Stock	12.02 (2.07)	11.87 (2.07)	10.51 (1.86)	-0.18 (0.19)	0.49*** (0.14)
Urban Share	6.72 (16.13)	9.22 (20.31)	1.40 (8.11)	-3.15 (2.56)	2.71** (1.15)
<i>Sectoral Employment Share</i>					
Agriculture	73.94 (21.3)	68.65 (23.51)	82.47 (20.39)	5.16 (3.22)	-1.57 (1.47)
Commerce	2.97 (5.42)	3.82 (5.22)	2.48 (5.03)	-0.69 (0.78)	0.09 (0.18)
Learned Professions and Engineers	2.25 (3.51)	4.20 (11.16)	2.00 (4.05)	-1.85 (1.42)	0.12 (0.12)
Manufacturing	17.38 (14.44)	19.51 (15.66)	10.53 (12.85)	-1.94 (1.66)	2.19* (1.31)

- ▶ Counties receiving normal schools are different than other counties
- ▶ but they are similar to counties receiving asylums

HBCU Balance in 1850

Table: County Characteristics in 1850

	(1) Black Normal	(2) Non-Black Normal	(3) Black Asylum	(4) Non-Black Asylum	(5) Within-State Difference (1) – (3)	(6) Within-State Difference (2) – (4)
Population, 1850	14,682 (8,809)	19,737 (29,806)	17,123 (9,012)	16,989 (16,378)	1,383.03 (5,054.12)	3,967.53 (18,605.03)
Proportion of population, 1850:						
Urban, Places 2,500 and over	0.09 (0.15)	0.07 (0.16)	0.22 (0.31)	0.13 (0.25)	-0.18 (0.17)	-0.12 (0.32)
In cities, 25,000 and over	0.01 (0.03)	0.01 (0.11)	0.00 (0.00)	0.02 (0.13)	0.02 (0.04)	0.06 (0.08)
Non-white, free	0.03 (0.04)	0.02 (0.03)	0.04 (0.06)	0.02 (0.04)	-0.025 (0.022)	-0.017 (0.013)
Non-white, slave	0.40 (0.18)	0.19 (0.20)	0.36 (0.16)	0.19 (0.18)	0.18** (0.06)	-0.07 (0.10)
Farmer	0.25 (0.08)	0.28 (0.12)	0.27 (0.19)	0.23 (0.11)	0.03 (0.11)	0.02 (0.13)
Real estate value per capita	320.60 (154.62)	216.26 (125.66)	266.74 (152.79)	296.50 (200.29)	145.93** (70.26)	-70.12 (66.54)
First Principal Component					0.06 (1.02)	-0.36 (1.58)

Notes: Columns (5), (6) ALL IK standard errors. For (1) and (3), sample size is 14 states with 25 counties. For (2) and (4), sample size is 14 states 60 counties.

Balance on Historical Mobility: Children in 1850

Table: Economic and Social Mobility for Children in 1850

	(1)	(2)	(3)	(4)	(5)	(6)
	Enrolled	Enrolled	Top Quartile Real Estate	Top Quartile Personal Estate	Married	Occ. Mobility
Year of Observation	1850	1850	1860	1860	1860	1860
Age in 1850	7-13	14-17	16-18	16-18	16-18	16-18
Grew up in Normal School County	-0.032 (.044)	-0.005 (.037)	0.008 (0.016)	0.005 (0.013)	0.005 (0.016)	-0.034* (0.017)
Observations	102	102	102	102	102	102
R-Squared	0.507	0.484	0.510	0.548	0.404	0.436
Mean DV, Asylum Counties	0.486	0.378	0.234	0.203	0.593	0.501
p-value randomization inference	0.474	0.899	0.58	0.725	0.761	0.091
Parental SES	Low	Low	Low	Low	Low	All

- No evidence of higher mobility for children in normal school v. asylum counties in the pre-period

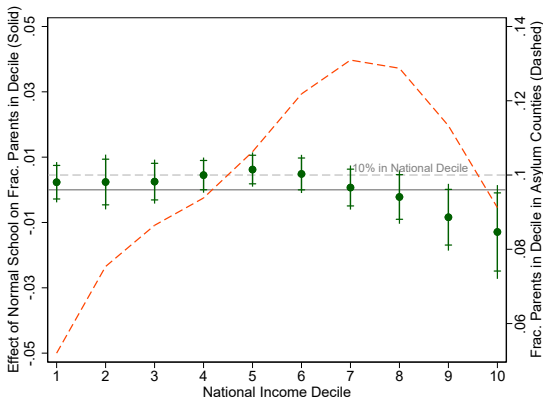
Effect on Social Mobility for Children in 1920

Table: Economic and Social Mobility for White Children in 1920, Measured in 1940

	(1)	(2)	(3)	(4)	(5)	(6)
	≥ HS	≥ Some College	≥ College	Married	Employed	Ln(HH Income)
Panel A: Males, Lower Socioeconomic Status						
Grew up in Normal School County	0.007 (0.007)	0.007 ⁺ (0.004)	0.002 (0.002)	0.005 (0.004)	-0.006 ⁺ (0.003)	-0.026 (0.016)
Observations	314	314	314	314	314	314
R-Squared	0.583	0.497	0.425	0.640	0.362	0.495
Mean DV, Asylum Counties	0.308	0.113	0.055	0.707	0.901	7.027
Panel B: Females, Lower Socioeconomic Status						
Grew up in Normal School County	0.010 (0.010)	0.024 ^{**} (0.006)	0.008 ^{**} (0.003)	0.019 [*] (0.008)	-0.020 ^{**} (0.007)	-0.013 (0.015)
Observations	314	314	314	314	314	314
R-Squared	0.474	0.408	0.355	0.675	0.657	0.471
Mean DV, Asylum Counties	0.404	0.140	0.047	0.701	0.297	7.052

- Teachers colleges were already ↑ education for local children in early 1900s
- 1920 county characteristics
- 1940 county characteristics
- No meaningful effects on Black children in the HBCU sample

Late 1990s: Regional Univ. Do Not Affect Parental Income Distribution



- ▶ Similar fraction of parents in each national income decile
 - ▶ Normal school counties have slightly higher frac. in middle and lower at top

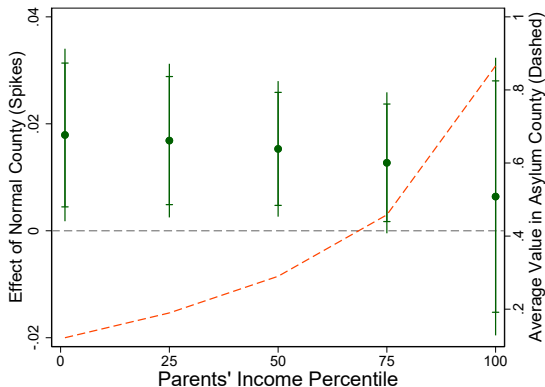
Effect on Higher Education Sector

Other county characteristics

Data on Economic and Social Mobility, 2005-2015

- ▶ Opportunity Insights data from Chetty et al. (2018)
 - ▶ IRS and Census data
 - ▶ County-level outcomes of children born between 1978-1983, who grew up in the county, by parental income
 - ▶ Includes 96% of all children born from 78-83, who were born in the US, or authorized immigrants who arrived in the US as children with parents who were citizens or authorized immigrants

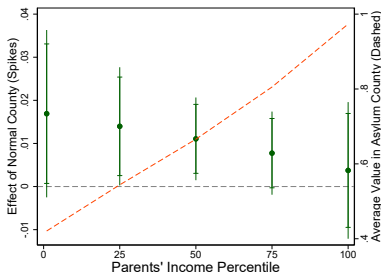
Regional Universities Raise College Attainment of Nearby Children



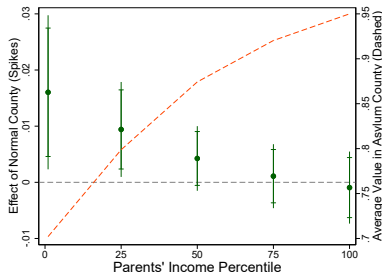
(a) At least 4-year College Degree, Age 25 and over

- ▶ Regional public universities raise the fraction of local children attaining a four-year degree by 2 percentage points
 - ▶ 8% for children with parental income at 25th percentile

Regional Universities $\uparrow$ Some College and HS Attainment of Nearby Children



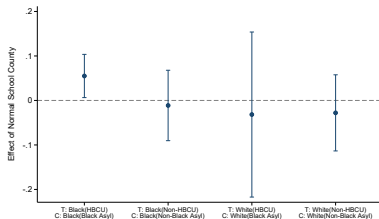
(a) At Least Some College, Age 25 and over



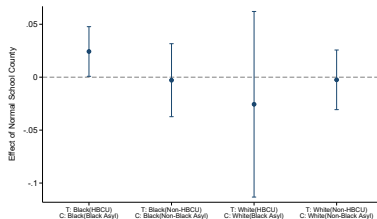
(b) At least HS Graduate or GED, Age 19 and over

Comparison to Causal Effects on Individuals

HBCUs Raise Educational Attainment of Nearby Black Children



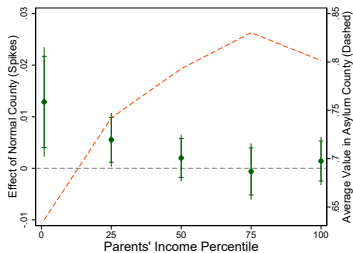
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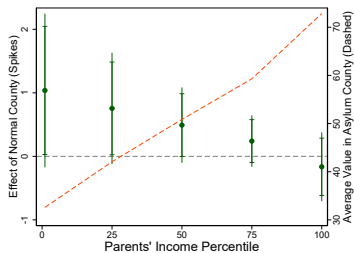
(b) At Least HS Graduate or GED

Figure: Effect of a Black Normal School on Educational Attainment, 2005-2015.

Regional Univ. Raise Employment and Income of Nearby Children



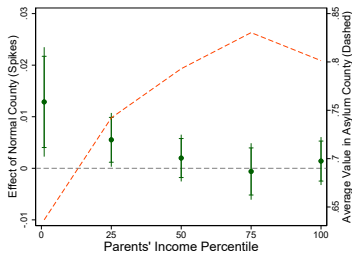
(a) Positive W-2 Earnings, 2015



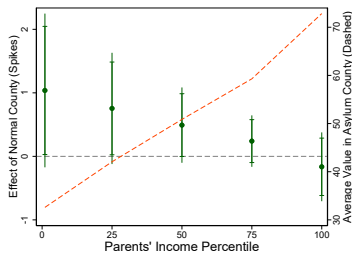
(b) Family Income Percentile, 2014-2015

- ▶ For children with parents at 25th income percentile
 - ▶ .6 p.p. $\uparrow$ in likelihood of W-2 earnings at age 32-37
 - ▶ Large effects on add'l degree recipients, if driven by education

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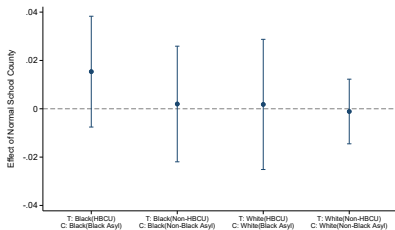
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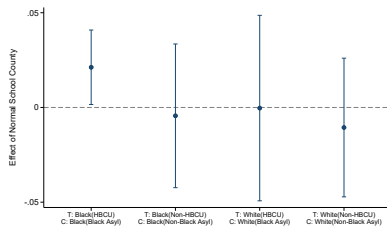
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 - ▶ .6 p.p. $\uparrow$ in likelihood of W-2 earnings at age 32-37
 - ▶ Large effects on add'l degree recipients, if driven by education
 - ▶ .7 pctile rank $\uparrow$ in HH income percentile
 - ▶ Chetty and Hendren (2018): 1 SD lower racial segregation as child $\Rightarrow$ 1.6 pctile higher HH rank (non-causal)

HBCUs Raise Employment and Income for Nearby Black Children



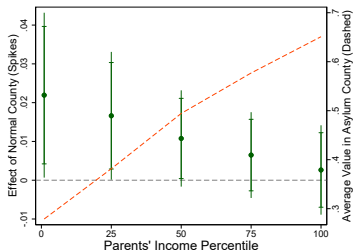
(a) Fraction with W-2 Earnings ≥ 0 , 2015



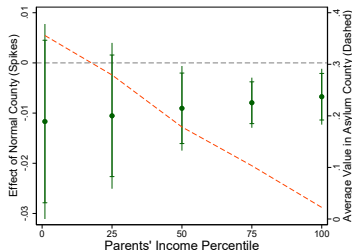
(b) Family Income Percentile, 2014-2015

Figure: Effect of a Black Normal School on Labor Market Outcomes, 2014-2015.

Regional Univ. Increase Marriage, Decrease Teen Births



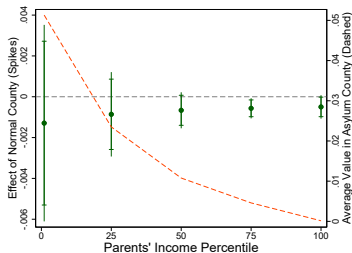
(a) Marriage, 2015



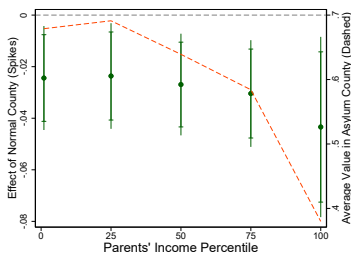
(b) Teen Birth (women-only)

- ▶ 1.5 p.p. (4%) increase in marriage, 25th pctl of family income
- ▶ 1 p.p. lower likelihood of teen birth

Regional Univ. Decrease Incarceration, Increase Geographic Mobility



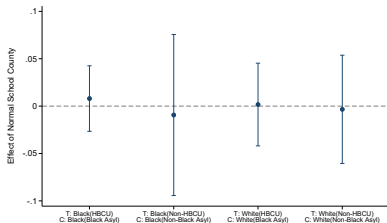
(a) Incarcerated, April 1, 2010



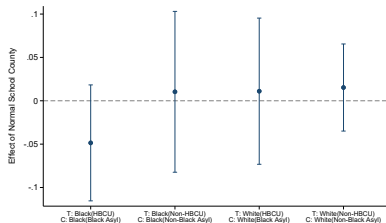
(b) Live in Childhood CZ

- For children with parents at 75th percentile of income distribution
 - .05 p.p. (12.5%) ↓ in incarceration
 - 3 p.p. (5%) ↓ likelihood of living in childhood CZ

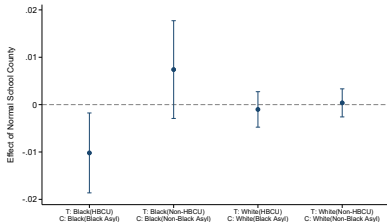
HBCUs Affect Social Outcomes of Black Children



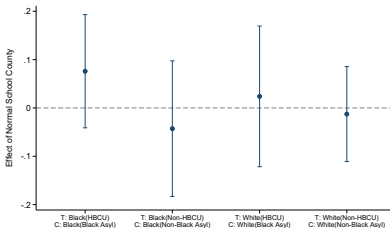
(a) Marriage, 2015



(b) Teen Birth (women-only)



(c) Incarcerated, April 1, 2020



(d) Living in Childhood CZ

Potential Channels

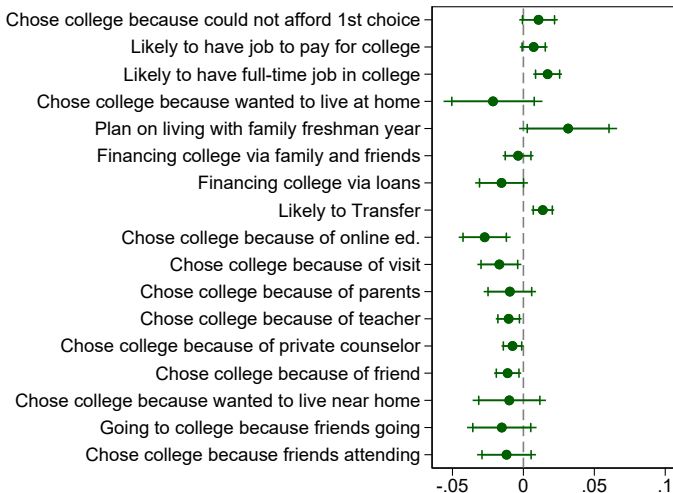
- ▶ Why do regional universities not reach children in asylum counties (as much)?
 - ▶ financial constraints (can't live at home)
 - ▶ information frictions
 - ▶ don't want to leave home
 - ▶ individual subsidies and information will be less effective

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 - ▶ financial constraints (can't live at home)
 - ▶ information frictions
 - ▶ don't want to leave home
 - ▶ individual subsidies and information will be less effective
- ▶ Restricted-access Freshman Survey data: 2.5 million college students who grew up in normal-school and asylum counties

$$y_i = \beta \text{Grew up in Normal School County}_i + \alpha_s + \epsilon_i$$

Proximity to a regional univ. lowers financial cost, information frictions



Robustness and Additional Exercises

- ▶ Historical Analysis
 - ▶ Representativeness of Census Tree linked records ●
 - ▶ Alternative mobility measure by parental occupation ●
- ▶ Opportunity Insights
 - ▶ Multiple hypothesis testing ●
 - ▶ Comparing close and far asylums ●
 - ▶ Heterogeneity by race and sex ●

Conclusion

- ▶ Regional universities improve local economic and social mobility
 - ▶ This was an important justification for their establishment
 - ▶ Larger effects for lower-income children
 - ▶ Causal effects on counties appear driven by causal effects on people, not sorting
 - ▶ New identification strategy based on placement of normal schools and insane asylums in the mid 1800s to early 1900s.

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 - ▶ Larger effects for lower-income children
 - ▶ Causal effects on counties appear driven by causal effects on people, not sorting
 - ▶ New identification strategy based on placement of normal schools and insane asylums in the mid 1800s to early 1900s.
- ▶ Regional universities induce local students on the margin, financially and academically
- ▶ Limited geographic reach due to financial costs, informational frictions

The County-Level Higher Education Sector in 1980

	(1) Variable Means	(2) Asylum	(3) Difference in Means With State FE (1) - (2)
Has regional college formerly normal school	0.91 (0.29)	0.00 (0.00)	0.93*** (0.02)
Total public four-year colleges	1.11 (0.67)	0.44 (0.88)	0.69*** (0.12)
Total private four-year colleges	1.39 (3.27)	1.94 (4.62)	-0.45 (0.53)
Total two-year colleges	0.97 (2.17)	1.16 (2.17)	-0.22 (0.31)
Enrollment as % of population	11.72 (9.23)	4.56 (5.51)	8.41*** (1.59)
Total degrees awarded as % of population	3.04 (2.77)	0.93 (1.41)	2.47*** (0.5)
% Population over 25 with Bachelor's degree	16.57 (4.79)	15.02 (6.1)	2.04** (0.86)

- ▶ Not all normal counties have a regional university
- ▶ Some asylum counties have a public four-year university
 - ▶ Reduced form will underestimate impact of regional university

The County-Level Higher Education Sector in 1980

	(1) Variable	(2) Means	(3) Difference in Means With State FE
	Normal	Asylum	(1) - (2)
Has regional college formerly normal school	0.91 (0.29)	0.00 (0.00)	0.93*** (0.02)
Total public four-year colleges	1.11 (0.67)	0.44 (0.88)	0.69*** (0.12)
Total private four-year colleges	1.39 (3.27)	1.94 (4.62)	-0.45 (0.53)
Total two-year colleges	0.97 (2.17)	1.16 (2.17)	-0.22 (0.31)
Enrollment as % of population	11.72 (9.23)	4.56 (5.51)	8.41*** (1.59)
Total degrees awarded as % of population	3.04 (2.77)	0.93 (1.41)	2.47*** (0.5)
% Population over 25 with Bachelor's degree	16.57 (4.79)	15.02 (6.1)	2.04** (0.86)

- ▶ Not all normal counties have a regional university
- ▶ Some asylum counties have a public four-year university
 - ▶ Reduced form will underestimate impact of regional university
- ▶ Asylum counties have more private and two-year colleges
 - ▶ But univ. in normal counties are public and much larger

Effect on the Higher Education Sector

	(1) Variable Means	(2) Asylum	(3) Difference in Means With State FE (1) - (2)
Has regional college formerly normal school	0.91 (0.28)	0.00 (00)	0.93** (0.02)
Total public four-year college	1.11 (0.67)	0.45 (0.88)	0.69** (0.13)
Total private four-year college	1.39 (3.27)	1.96 (4.63)	-0.48 (0.55)
Total two-year college	0.97 (2.17)	1.17 (2.17)	-0.24 (0.31)
Enrollment as % of population	11.72 (9.23)	4.60 (5.52)	8.48** (1.61)
Full-time enrollment as % population	8.52 (7.4)	2.99 (4.35)	6.51** (1.27)
Total degree awarded as % of population	3.04 (2.77)	0.94 (1.41)	2.48** (0.5)
Bachelor's degree awarded as % of population	1.43 (1.38)	0.39 (0.69)	1.23** (0.25)
% Population over 25 with Bachelor's degree	16.57 (4.79)	15.05 (6.11)	2.04* (0.86)
% population over 25 with 1-3 years college	15.40 (3.89)	15.03 (3.98)	0.55 (0.35)

Characteristics of Four-Year Universities

	Normal	Asylum	Diff Normal-Asylum
Public university	0.83 [.26]	0.4 [.39]	0.439*** (0.053)
Tuition and fees	9767 [6147]	17350 [9241]	-0.532*** (0.059)
Grant and scholarship aid	6227 [3867]	9984 [5573]	-0.485*** (0.068)
Net price, students receiving grant or scholarship aid	12748 [3977]	17021 [4794]	-0.246*** (0.042)
Grant and scholarship aid in income level (30,001-48,000)	8377 [5066]	13110 [7497]	-0.428*** (0.064)
Net price in income level (30,001-48,000), students receiving Title IV federal financial aid	11456 [3703]	14214 [3787]	-0.175*** (0.043)
BA completion within 150% normal time	0.5 [.15]	0.56 [.18]	-0.042** (0.020)
Open admission policy	0.09 [.27]	0.13 [.26]	-0.051 (0.035)
Conditional on non-open admission			
Fraction admitted	0.68 [.15]	0.68 [.14]	-0.002 (0.019)
Letters of recommendations required or recommended	0.46 [.46]	0.67 [.4]	-0.224*** (0.065)
Test scores required or recommended	0.97 [.07]	0.87 [.27]	0.111*** (0.031)
Conditional on non-open admission, test scores required, and $\geq 60\%$ of enrolled students submitted			
ACT Composite, 75th percentile	24.41 [2.61]	26.31 [2.4]	-1.800*** (0.268)
SAT Verbal, 75th percentile	568.37 [51.69]	598.41 [49.68]	-25.325*** (7.191)
SAT Math, 75th percentile	576.77 [55.98]	608.28 [57.37]	-23.838*** (8.454)

Back to Effects on Parental Income Distribution

Non-random selection into matched Census Tree records, 1850

DV = Linked Sample	Low SES	High SES
Male	0.183*** (0.010)	0.147*** (0.007)
Parental occupation score	0.000 (0.000)	-0.001*** (0.000)
Mother's age	0.002*** (0.000)	0.002*** (0.000)
Father's age	-0.000 (0.000)	0.000 (0.000)
Father head	-0.026 (0.022)	0.018 (0.044)
Mother head	-0.043*** (0.013)	-0.068 (0.050)
Lives with father	0.039 (0.027)	-0.043 (0.049)
Lives with mother	-0.077*** (0.020)	-0.045* (0.024)
Lives in birth state	0.047*** (0.010)	0.046*** (0.010)
Father lives in birth state	0.019*** (0.007)	0.006 (0.006)
Mother lives in birth state	0.002 (0.007)	0.007 (0.007)
Foreign born	0.066*** (0.014)	-0.013 (0.019)
Father foreign born	-0.033** (0.013)	-0.055*** (0.016)
Mother foreign born	-0.052*** (0.011)	-0.034** (0.017)
Number of siblings	0.012*** (0.001)	0.008*** (0.001)
First family unit in household	0.097*** (0.017)	0.113 (0.069)
Observations	51,164	80,369
R-squared	0.071	0.055
DV mean	0.424	0.544

Historical Mobility Adjusting for Representativeness

Table: Economic and Social Mobility for Children in 1850, with Propensity Score Weights

	(1)	(2)	(3)	(4)	(5)	(6)
	Enrolled	Enrolled	Top Quartile Real Estate	Top Quartile Personal Estate	Married	Occ. Mobility
Year of Observation	1850	1850	1860	1860	1860	1860
Age in 1850	7-13	14-17	16-18	16-18	16-18	16-18
Grew up in Normal School County	-0.032 (.044)	-0.005 (.037)	0.004 (0.016)	0.004 (0.013)	0.003 (0.016)	-0.035** (0.017)
Observations	102	102	102	102	102	102
R-Squared	0.507	0.484	0.505	0.533	0.425	0.427
Mean DV, Asylum Counties	0.486	0.378	0.247	0.209	0.599	0.504
p-value randomization inference	0.474	0.899	0.809	0.775	0.851	0.078
Parental SES	Low	Low	Low	Low	Low	All

Non-random selection into matched Census Tree records, 1920-1940

	(1)	(2)	(3)	(4)
Parental occupation score	-0.001*** (0.000)	0.001*** (0.000)	-0.001*** (0.000)	0.001*** (0.000)
Mother's age	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.003*** (0.000)
Father's age	-0.000** (0.000)	-0.000*** (0.000)	0.000 (0.000)	0.000 (0.000)
Father head	0.023*** (0.005)	0.018*** (0.003)	0.027*** (0.005)	0.024*** (0.004)
Mother head	0.018*** (0.003)	-0.010* (0.006)	0.011*** (0.003)	0.005 (0.004)
Lives with father	0.024*** (0.006)	0.034*** (0.007)	0.011* (0.007)	0.040*** (0.007)
Lives with mother	-0.061*** (0.006)	-0.050*** (0.006)	-0.073*** (0.010)	-0.076*** (0.011)
Lives in birth state	0.048*** (0.004)	0.047*** (0.002)	0.035*** (0.003)	0.036*** (0.003)
Father lives in birth state	0.018*** (0.003)	0.015*** (0.002)	0.019*** (0.004)	0.013*** (0.003)
Mother lives in birth state	0.016*** (0.003)	0.010*** (0.001)	0.010** (0.004)	0.007** (0.003)
Foreign born	-0.061*** (0.008)	-0.038*** (0.006)	-0.078*** (0.007)	-0.066*** (0.003)
Father foreign born	-0.050*** (0.003)	-0.039*** (0.002)	-0.066*** (0.004)	-0.052*** (0.003)
Mother foreign born	-0.059*** (0.004)	-0.050*** (0.002)	-0.076*** (0.005)	-0.062*** (0.003)
Number of siblings	0.005*** (0.000)	0.002*** (0.000)	0.005*** (0.000)	0.000 (0.001)
First family unit in household	0.111*** (0.007)	0.115*** (0.007)	0.097*** (0.005)	0.099*** (0.006)
Sample	Low SES Males	High SES Males	Low SES Females	High SES Females
Observations	1,278,770	1,934,526	1,256,846	1,925,246
R-squared	0.040	0.026	0.076	0.050
DV mean	0.666	0.681	0.398	0.390

Table: 1940 Outcomes of 1920 6-15 Year-Old White Children, County-Level Outcomes based on 1920 County, with Propensity Score Weights

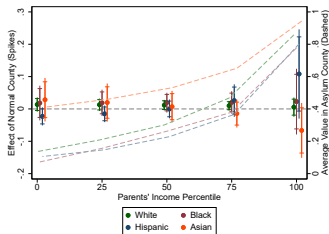
	(1)	(2)	(3)	(4)	(5)	(6)
	≥ HS	≥ Some College	≥ College	Married	Employed	Ln(HH Income)
Panel A: Males, Lower Socioeconomic Status						
Grew up in Normal School County	0.007 (0.007)	0.007* (0.004)	0.001 (0.002)	0.005 (0.004)	-0.006* (0.003)	-0.026 (0.016)
Observations	314	314	314	314	314	314
R-Squared	0.575	0.482	0.409	0.636	0.367	0.489
Mean DV, Asylum Counties	0.307	0.113	0.055	0.705	0.900	7.028
Panel B: Females, Lower Socioeconomic Status						
Grew up in Normal School County	0.008 (0.010)	0.023*** (0.006)	0.008*** (0.003)	0.020** (0.008)	-0.022*** (0.007)	-0.012 (0.015)
Observations	314	314	314	314	314	314
R-Squared	0.452	0.395	0.329	0.678	0.668	0.451
Mean DV, Asylum Counties	0.399	0.137	0.046	0.692	0.305	7.049

Alternative Mobility Measure 1850: Occupational Rank

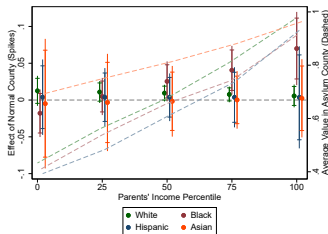
Table: Sons' predicted occupational score percentile in 1860 conditional on father's in 1850. White children

	(1)	(2)	(3)
Grew up in Normal School County	-0.009 (0.013)	-0.000 (0.013)	0.009 (0.023)
Observations	102	102	102
R-squared	0.366	0.310	0.246
Mean DV, Asylum Counties	0.420	0.474	0.528
p-value randomization inference	0.52	0.998	0.748
Father's Occupation Score Percentile	25	50	75

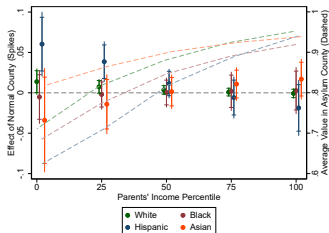
Education Results by Race



(a) $\geq$ 4-year Degree, Age 25 and over

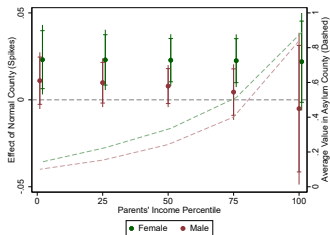


(b) $\geq$ Some College, Age 25 and over

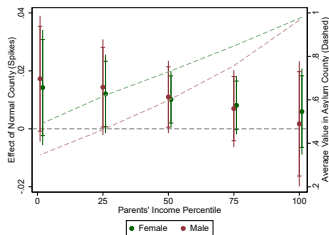


(c) $\geq$ HS Grad or GED, Age 19 and over

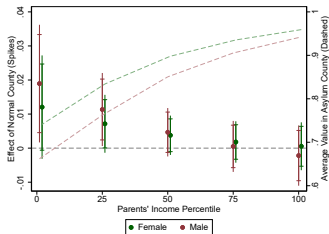
Education Results by Sex



(a) $\geq$ 4-year Degree, Age 25 and over



(b) $\geq$ Some College, Age 25 and over

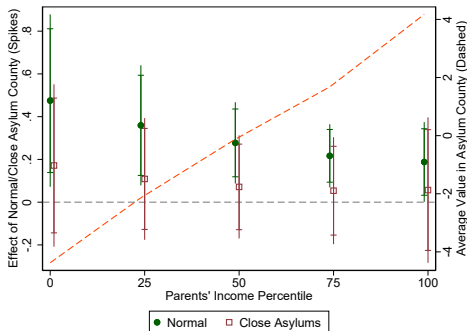


(c) $\geq$ HS Grad or GED, Age 19 and over

Comparing Close and Far Asylums

$$y_i = \beta_{\text{normal}} \text{Normal}_i + \beta_{\text{close asylum}} \text{Close Asylum}_i + \alpha_s + \epsilon_i$$

- ▶ β_{normal} : normal school v. far-away asylum (> 41 miles)
- ▶ $\beta_{\text{close asylum}}$: close asylum v. far-away asylum counties



- ▶ Cannot rule out zero or substantial spillovers to close asylum counties

Comparison to Causal Effects of Regional Univ. on People

- ▶ Previous figure: causal effects of regional univ. on counties
- ▶ Do not identify causal impact on the child
 - ▶ Universities may affect composition of who grows up in the county
- ▶ Use causal effects of counties on mobility from Chetty and Hendren (2018)

Table: Causal Effects on College Attendance, 25th percentile parental income

	(1) Some College Age 25+	(2) Attended College Age 18-23
Normal	1.397* (0.680)	0.143+ (0.0764)
Observations	324	305
Weights	Unweighted	Precision Weights
Scale	Per Childhood	Per Year
Interpretation	Effect on Place	Effect on Person

- ▶ Similar magnitude for causal effect on county and on individuals

Comparison to Causal Effects of Regional Univ. on People

Table: Causal Effects on Income, 25th percentile parental income

	(1)	(2)
	Family Income Pctile, 2014-15	Family Income Pctile, Age 26
Normal	0.748 ⁺	0.0794 ⁺
	(0.428)	(0.0428)
Observations	325	306
Birth Cohorts	1978-1983	1980-1986
Weights	Unweighted	Precision Weights
Scale	Per Childhood	Per Year
Interpretation	Effect on Place	Effect on Person

- ▶ Comparable causal effects on county and on individuals

Comparison to Causal Effects of Regional Univ. on People

Table: Causal Effects on Marriage, 25th percentile parental income

	(1) Married 2015	(2) Married Age 26
Normal	1.662* (0.817)	0.0856+ (0.0470)
Observations	324	300
Weights	Unweighted	Precision Weights
Scale	Per Childhood	Per Year
Interpretation	Effect on Place	Effect on Person

- ▶ Comparable causal effects on county and on individuals
- ▶ Causal effects on individuals not available for other social outcomes

[Back to Effects on Place](#)

Students from normal school counties more likely attend college near home

	(1) Attend Univ. within 10mi	(2) Attend Former-Normal within 10mi	(3) Attend Univ in county	(4) Attend Former-Normal in county
Grew up in normal school county	0.0462** (0.0131)	0.0787*** (0.0127)	0.149*** (0.0313)	0.202*** (0.0297)
Observations	324	324	324	324

Standard errors clustered by state. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Observations are at the county level. All regressions include state fixed effects.

- ▶ Driven by enrollment at previous normal school

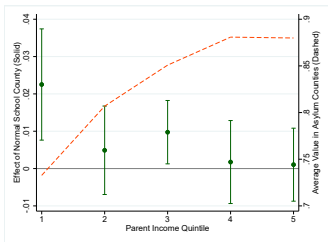
Balance in 1920

	(1)	(2)	(3)	(4)	(5)
	Variable Means			Difference in Means With State FE	
	Normal	Asylum	All others	(1) - (2)	(1) - (3)
Log Population	10.66 (1.102)	10.86 (1.154)	9.65 (0.946)	-0.14 (0.153)	0.69*** (0.091)
Log Nearby Population	11.91 (0.473)	11.97 (0.438)	11.81 (0.383)	-0.042* (0.023)	0.01 (0.01)
Urban Share	39.97 (28.395)	46.60 (26.917)	16.05 (22.389)	-3.02 (3.929)	18.086*** (2.572)
Male Share	50.98 (1.838)	51.35 (2.578)	51.95 (2.38)	-0.26 (0.283)	-0.894*** (0.176)
African-American Share	10.14 (18.186)	7.01 (13.671)	12.12 (19.643)	0.47 (0.607)	2.154*** (0.703)
White Foreign-Born Share	10.18 (9.402)	11.18 (8.573)	6.38 (7.719)	-1.228* (0.665)	0.946*** (0.356)
Population Density	0.37 (1.762)	1.27 (9.574)	0.07 (0.541)	-0.99 (1.179)	0.17 (0.113)
Log Manufacturing Establishments	4.63 (1.333)	4.86 (1.37)	3.45 (1.027)	-0.18 (0.17)	0.737*** (0.084)
Average Manufacturing Wage	1,035.13 (231.487)	1,071.53 (252.168)	972.40 (284.351)	-7.58 (27.886)	24.73 (15.754)
Manufacturing Employment per Population	6.88 (6.586)	7.78 (6.202)	3.62 (5.288)	-0.968* (0.575)	1.271*** (0.334)
Log Value Added, Manufacturing	15.18 (2.133)	15.58 (2.156)	13.26 (1.944)	-0.32 (0.243)	1.211*** (0.125)
Log Value of Crops	15.30 (1.21)	15.42 (0.997)	14.87 (1.19)	0.04 (0.141)	0.391*** (0.097)
Log Value of Farm Property	16.94 (1.085)	17.15 (0.962)	16.49 (1.084)	0.04 (0.117)	0.437*** (0.083)

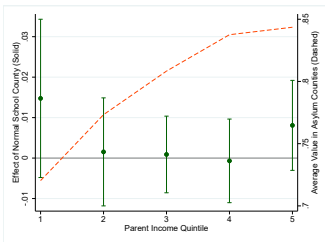
Current County Characteristics from Opportunity Insights

	Normal	Asylum	Within-State Difference
Economic Characteristics			
Manufacturing employment share, 2000	0.13 [.06]	0.15 [.07]	-0.01 (0.01)
Average annualized job growth, 2004-2013	0 [.01]	0 [.01]	0.00 (0.00)
HS grad. wage growth, 2005-2009 - 2010-2014	0.06 [.11]	0.05 [.07]	0.01 (0.01)
Bachelor's degree share, age ≥ 25, 2000	0.24 [.07]	0.22 [.09]	0.03* (0.01)
Population, 2000	269,614 [765,738]	305,908 [593,196]	-30,670.68 (92,294)
Children < 18, 2000	67,974 [209,691]	77,295 [152,889]	-8,224.44 (24,235)
K-12 Public Schools and Colleges			
K-12 expenditures per stud., 1996-1997	6.38 [1.43]	6.38 [1.41]	-0.00 (0.07)
K-12 student teacher ratio, 1996-1997	16.88 [2.18]	17.52 [2.13]	-0.46* (0.18)
Mean 3rd grade math test scores, 2013	3.28 [.63]	3.29 [.71]	0.03 (0.07)
College tuition, local colleges, IPEDS 2000	4149.01 [3,836.2]	6778.76 [4,664.33]	-2,455.74** (589.25)
Family characteristics, children in Chetty et al. (2018)			
Children claimed by two people			
parent income at p25	0.51 [.12]	0.49 [.12]	0.03* (0.01)
parent income at p75	0.94 [.04]	0.93 [.06]	0.00 (0.01)
Fraction of childhood spent in the county	0.74 [.07]	0.76 [.06]	-0.01+ (0.01)

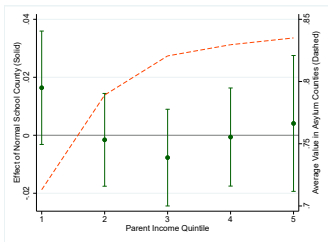
Employment Results by Family Composition



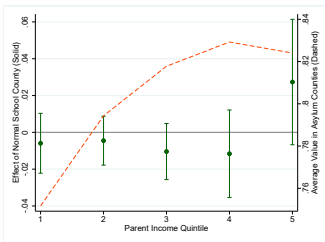
(a) Men, two parents



(b) Women, two parents



(c) Men, single parent



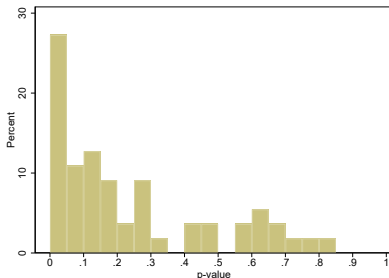
(d) Women, single parent

Multiple Hypothesis Testing

- ▶ Investigate 11 outcomes at 5 parental income percentiles
- ▶ Are results robust to considering multiple hypothesis testing?

Multiple Hypothesis Testing

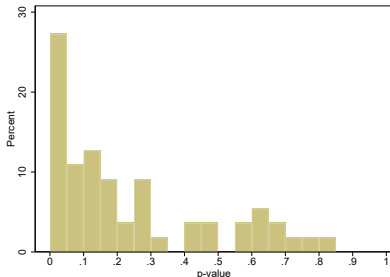
- ▶ Investigate 11 outcomes at 5 parental income percentiles
- ▶ Are results robust to considering multiple hypothesis testing?



- ▶ More than 25% of p-values < 5%

Multiple Hypothesis Testing

- ▶ Investigate 11 outcomes at 5 parental income percentiles
- ▶ Are results robust to considering multiple hypothesis testing?



- ▶ More than 25% of p-values $< 5\%$
- ▶ Implement Young (2020), reject null that there is no effect on the 55 outcomes

Multiple Hypothesis Testing: Principal Components

- ▶ Want to test for effect on “social mobility”, common part of each outcome
 - ▶ and particularly for lower-income children
- ▶ Calculate principal component of the 11 outcomes
- ▶ Test for effect at each of the five percentiles
- ▶ Adjust for 5 different tests using Romano and Wolf (2005)
 - ▶ Adjusted p-values under 5%, except for 100th ($p = .12$)

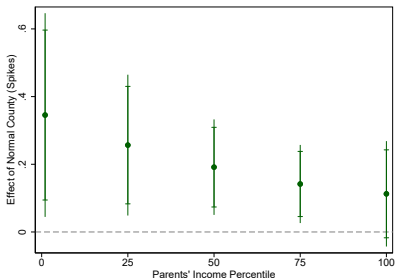


Figure: Principal Components

Takeaways

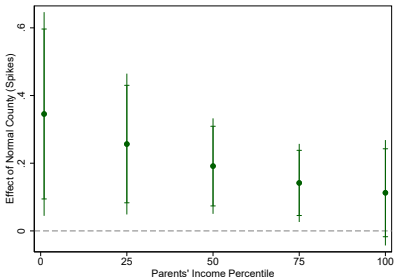


Figure: Principal Components

- ▶ Larger point estimates for lower-income children
- ▶ These differences not a key aspect of our study
- ▶ If regional univ. help low-income children, $\uparrow$ relative to nat'l distribution
 - ▶ Regardless of whether they help high-income children

Takeaways

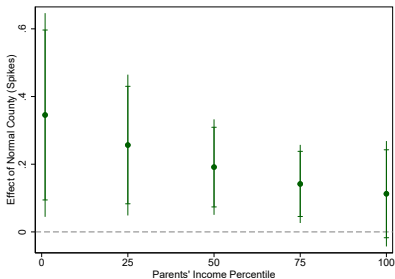


Figure: Principal Components

- ▶ Larger point estimates for lower-income children
- ▶ These differences not a key aspect of our study
- ▶ If regional univ. help low-income children, ↑ relative to nat'l distribution
 - ▶ Regardless of whether they help high-income children
- ▶ College completion rates only 50% for children in asylum counties with parents at 75th pctile (\$95,000)

Back to robustness

Population, Labor Force, and Dwelling Characteristics in 1940

	Normal	Asylum	Within-State Difference
Population	121743.01 [263443.58]	174916.86 [474510.93]	-38,335.178 (44,501.932)
Fraction of Total Population			
Urban	0.45 [.27]	0.51 [.23]	-0.030 (0.028)
Male	0.5 [.02]	0.51 [.02]	-0.001 (0.002)
Black	0.09 [.16]	0.07 [.12]	-0.000 (0.010)
Foreign	0.06 [.06]	0.07 [.06]	-0.011** (0.004)
Ln(Average Wage, Manufacturing)	6.83 [.29]	6.87 [.28]	0.019 (0.026)
Ln(Value Added, Manufacturing)	15.22 [2.22]	15.5 [2.22]	-0.241 (0.213)
Ln(Manufacturing Establishments)	3.93 [1.56]	4.24 [1.53]	-0.297 ⁺ (0.152)
Fraction Employed, Age 14 and Over	0.86 [.05]	0.86 [.05]	-0.004 (0.006)
Fraction Seeking Work, Age 14 and Over	0.09 [.03]	0.08 [.04]	-0.001 (0.004)
Fraction of Occupied Dwelling Units Occupied by Owners	0.48 [.11]	0.49 [.1]	0.002 (0.010)
Fraction of Occupied Dwelling Units with			
Electric Lighting Equipment	0.71 [.23]	0.77 [.19]	-0.022 (0.018)
Mechanical Refrigeration Equipment	0.36 [.16]	0.41 [.15]	-0.020 (0.015)
Ln(Average Value of Owner Occupied Housing)	7.9 [.4]	7.98 [.42]	-0.050 (0.038)
Ln(Median Value of Owner Occupied Housing)	7.65 [.53]	7.79 [.48]	-0.076 ⁺ (0.045)
Ln(Value of Crops Harvested)	14.82 [.96]	14.87 [.81]	0.013 (0.107)
First Principal Component	-0.18 [2.94]	0.4 [2.72]	-0.325 (0.261)

Occupation in 1940

	Normal	Asylum	Within-State Difference
Fraction of Employment by Occupation			
Professional	0.07 [.01]	0.07 [.02]	-0.0004 (0.002)
Semiprofessional	0.01 [.004]	0.01 [.004]	-0.0003 (0.0004)
Farmers and farm-managers	0.16 [.13]	0.14 [.12]	0.025* (0.012)
Proprietors, managers, and officials (except farm)	0.08 [.02]	0.08 [.02]	0.002 (0.002)
Clerical, sales, and kindred workers	0.13 [.06]	0.14 [.06]	-0.005 (0.006)
Craftsmen, foremen, and kindred workers	0.1 [.04]	0.11 [.04]	-0.004 (0.004)
Operatives and kindred workers	0.16 [.09]	0.16 [.08]	-0.003 (0.009)
Domestic service	0.05 [.03]	0.05 [.02]	-0.002 (0.002)
Service, except domestic	0.07 [.03]	0.08 [.03]	-0.016** (0.004)
Laborer (except farm)	0.07 [.04]	0.07 [.04]	-0.005 (0.004)

Industry in 1940

	Normal	Asylum	Within-State Difference
Fraction of Employment by Industry			
Agriculture, Forestry, and Fishing	0.25	0.22	0.028
	[.17]	[.17]	(0.020)
Mining	0.02	0.01	0.009*
	[.05]	[.03]	(0.004)
Construction	0.09	0.09	0.001
	[.03]	[.02]	(0.003)
Manufacturing	0.18	0.2	-0.024*
	[.11]	[.12]	(0.010)
Transportation, Communication, and Other Utilities	0.06	0.06	0.002
	[.03]	[.03]	(0.003)
Wholesale and Retail Trade	0.14	0.15	-0.001
	[.04]	[.03]	(0.005)
Finance, Insurance, and Real Estate	0.02	0.02	-0.001
	[.01]	[.01]	(0.002)
Business and Repair Services	0.02	0.02	0.0004
	[.004]	[.003]	-0.0004
Personal Services	0.08	0.08	-0.003
	[.03]	[.03]	(0.003)
Professional and Related Services	0.08	0.08	-0.007**
	[.02]	[.03]	(0.002)
Public Administration	0.03	0.04	-0.004
	[.03]	[.04]	(0.003)
First Principal Component	-0.29	0.47	-0.389
	[2.88]	[2.67]	(0.345)

School Attendance and Educational Attainment in 1940

	Normal	Asylum	Within-State Difference
Fraction Attending School by Age			
5 to 6	0.42 [.16]	0.44 [.15]	0.001 (0.009)
7 to 13	0.96 [.07]	0.96 [.04]	-0.001 (0.005)
14 to 15	0.9 [.07]	0.91 [.06]	0.004 (0.006)
16 to 17	0.69 [.11]	0.7 [.1]	0.016* (0.008)
18 to 20	0.26 [.07]	0.25 [.06]	0.025** (0.006)
21 to 24	0.06 [.03]	0.06 [.03]	0.007* (0.003)
Fraction of 16 to 17 year-olds			
Attending school, completed HS	0.05 [.02]	0.04 [.02]	0.007* (0.003)
Attending school, did not yet complete HS	0.63 [.11]	0.65 [.1]	0.010 (0.007)
Median School Years Completed, Age 25 and Over			
Male	8.09 [.93]	8.27 [.72]	0.041 (0.071)
Female	8.65 [1.08]	8.75 [.88]	0.160+ (0.085)
Fraction Age 25 and Over with			
Four or more years of college	0.05 [.01]	0.04 [.02]	0.005* (0.002)
One to three years of college	0.15 [.04]	0.15 [.04]	0.009** (0.003)
Four years of high school	0.13 [.04]	0.14 [.04]	0.001 (0.004)

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